

MuscleSim — from a shock to a twitch

Group worksheet foxworthywa.github.io/neuronsim/muscle.html

Names: _____

The whole story in one line: a command arrives at the end plate → an impulse runs over the sarcolemma and down the T-tubules → the SR lets Ca^{2+} out → Ca^{2+} lets the myofibrils pull → the SR pumps Ca^{2+} back and the fibre relaxes. Keep a finger on this line; every scene is one step of it.

How to use this sheet. Part A before you open the sim. In Part B the numbers match the circles across the top of the sim: when a box names a scene, stop there and answer together before pressing Continue.

TALK FIRST means every person says their answer out loud before anyone writes — disagree first, then write what you agreed. One sheet per group.

A · Before you open the sim from the introduction, and what you already know

Draw a resting muscle fibre. The ions are Na^+ , K^+ , Ca^{2+} , Cl^- , and the large proteins (A^-) that are stuck inside. Write each one on the side where it is more concentrated (bigger letters = more of it). Mark the inside + or –.

outside the fibre

TALK FIRST If only K^+ channels open, which way does K^+ move — in or out? Does the inside become more + or more –? If only Na^+ channels open?

TALK FIRST The Na^+/K^+ pump uses ATP to push Na^+ out and pull K^+ in. Is its job to make the charge right now, or to keep the gradients that the channels use? What would your drawing look like if the pump had been off for an hour?

B · Working through the sim

2 Parts of a muscle fibre · on screen: “Find the structure”

For each part, its job in a few words (what it is for, not what it looks like), then number them in the order the signal reaches them.

Structure	Its job	Order (1-5)
Sarcolemma		
T-tubules		
Sarcoplasmic reticulum (cisternae)		

Myofibrils		
Motor end plate		

3 The resting membrane · on screen: "the concentration table on screen"

Check your Part A drawing against the concentrations on screen. Correct it in a different colour. What, if anything, did you have the wrong way round? _____

5 Threshold and all-or-none · on screen: "Channels with a voltage sensor · Double the shock"

Shocks of 20, 40, 60 and 80 % each pushed V_m up, and each time it fell straight back. The 100 % shock fired a full impulse. V_m at the moment the sim paused — when the first Na^+ channels opened — is the threshold: _____ mV.

TALK FIRST What was the membrane doing at 100 % that it was not doing at 80 %? (Hint: what does Na^+ coming in do to the other Na^+ channels?)

Complete the loop: Na^+ channels open $\rightarrow \text{Na}^+$ comes _____ $\rightarrow$ the inside becomes more _____ $\rightarrow$ _____
 Na^+ channels open $\rightarrow$... What pulls V_m back down after a small shock, before this loop can get going?

Shock at 100 % and at 200 %: the two impulses were (circle one) the same size / different. Why does a bigger shock not make a bigger impulse?

TALK FIRST Which everyday thing is threshold most like — a dimmer switch, a sneeze, or a ball pushed up a hill? Say why. In the fibre, what is the push, and what is the falling back?

6 Building the action potential · on screen: "Predict: what happens next? ... Two shocks"

The sim pauses at each phase. Fill in the table as it stops.

Phase	Which gated channel is open	Which ion moves, and which way
Rising phase		
At the peak		
Falling phase		
Undershoot		

TALK FIRST Two shocks 3 ms apart: the second shock produced _____. What state were the Na^+ channels in?

7 Along the fibre and down the tubes · on screen: "Shock the middle of the fibre"

TALK FIRST The impulse ran to both ends. Why does it never turn round and travel back the way it came?

8 Voltage to calcium · on screen: "The classic experiment: no Ca^{2+} outside"

TALK FIRST VOTE before you run the Ca^{2+} -free bath — with no Ca^{2+} outside the fibre, will it still twitch? Yes ☐ ☐ ☐ ☐ ☐ No ☐ ☐ ☐ ☐ ☐ (one tick per person)

Result (circle one): full twitch / weaker twitch / nothing. So where does the Ca^{2+} for a contraction come from? _____

9-10 Calcium to force, and letting go · on screen: "Letting go · No ATP"

When someone dies, their cells stop making ATP. Using this scene, explain why the body stiffens a few hours later — and why it goes stiff rather than limp.

TALK FIRST Is a stiff muscle after death contracting? What is it actually doing? (Two things need ATP here; name both.)

11 One twitch, phase by phase · on screen: *"The same twitch, one phase at a time"*

The sim stops at each phase. Number these 1-8 in the order they happen:

___ Ca ²⁺ floods out of the store	___ SERCA pumps Ca ²⁺ back into the store
___ Relaxed	___ Troponin catches Ca ²⁺ , tropomyosin moves
___ The impulse: Na ⁺ channels open	___ The impulse runs down the T-tubules
___ Cross-bridges cycle; force rises	___ Peak force

TALK FIRST Which lasts longest (circle one): the impulse / the Ca²⁺ / the force. Why does that gap matter for the next scene?

12 Summation and tetanus · on screen: *"Summation · Raise the rate"*

Two shocks 20 ms apart. The second impulse was (circle) bigger / smaller / the same. The force was (circle) bigger / smaller / the same.

TALK FIRST So what adds up, and what does not? When you raised the rate until the force went smooth, what ran out of time between impulses?

When you hold a cup steady, are the fibres in your arm twitching, or in tetanus? _____
(needed in Part C)

13 The neuromuscular junction · on screen: *"The end-plate potential by itself · Block half"*

With the Na⁺ channels switched off, the receptors alone pushed V_m (circle one) short of threshold / just to threshold / well past threshold.

TALK FIRST With half the receptors blocked, the twitch was (circle one) smaller / the same. So what would have to happen for a command from the nerve to fail?

C · After the sim: your own muscle the EMG demo — write each prediction before you look

Look back at your cup answer in scene 12. Your instructor will contract a muscle once, briefly. The sim showed one impulse for one twitch. How many impulses will the EMG show — one, or many? _____

Gentle squeeze, then hard squeeze. Will the signal get taller, busier, or both? _____

Afterwards: what did you actually see, and which scene of the sim explains it?

Instructor key

Answers in the terms the sim uses. The one number asked for is threshold; other model values are given only where a circle needs the fact behind it.

A Outside: Na^+ , Ca^{2+} , Cl^- high. Inside: K^+ high, A^- trapped, inside negative (≈ -85 mV). K^+ channels open $\rightarrow \text{K}^+$ out $\rightarrow$ inside more negative. Na^+ channels open $\rightarrow \text{Na}^+$ in $\rightarrow$ inside more positive. The pump keeps the gradients; it does not make the charge in the moment — switch it off and V_m barely changes for a long time (the sim lets them try this in the Lab). Off for an hour: the gradients run down, the drawing goes flat, nothing can fire.

2 Sarcolemma carries the impulse over the surface; T-tubules carry it inward; SR (its cisternae beside each tubule) stores and releases Ca^{2+} ; myofibrils pull; the end plate is where the nerve delivers its command. Order: end plate 1, sarcolemma 2, T-tubules 3, SR 4, myofibrils 5.

3 Na^+ 145 out / 12 in; K^+ 4 out / 140 in; Cl^- 110 out / 6 in; Ca^{2+} 2 out / 0.0001 in (mM). The commonest error is K^+ on the wrong side.

5 Threshold ≈ -60 mV. At 80 % a few Na^+ channels open but the K^+ leak pulls V_m back before more can join; at 100 % enough open that the Na^+ coming in depolarizes the membrane further and opens more — the loop outruns the leak. Loop: in $\rightarrow$ positive $\rightarrow$ more. The K^+ leak is what pulls it back. 100 % vs 200 %: the same size (both $\approx +49$ mV) — once the loop has started the channels do the work, not the shock; the shock's only job was getting V_m to threshold. Best analogy: a sneeze or a ball pushed over a hill (a dimmer is graded, which is exactly what an impulse is not). The push is the shock; the falling back is the K^+ leak.

6 Rising: voltage-gated Na^+ open, Na^+ in. Peak: Na^+ inactivating, K^+ opening. Falling: K^+ open, K^+ out. Undershoot: K^+ still open. Second shock at 3 ms: nothing — Na^+ channels inactivated (refractory).

7 Behind the wave the Na^+ channels are inactivated, so the membrane it has just left cannot be re-excited until they reset — by then the wave is gone.

8 Full twitch (the model gives 0.27, identical to normal). All the Ca^{2+} for contraction comes from the SR; none from outside. (In the sim, Ca^{2+} outside matters at the nerve terminal, scene 13, not in the fibre.)

9-10 Rigor mortis. ATP is needed (1) for myosin heads to let go of actin and re-cock, and (2) for SERCA to pump Ca^{2+} back into the store. With no ATP the leaked Ca^{2+} is never removed and the bridges that form can never release, so force climbs and stays: stiff, not limp. It is not contracting — nothing is cycling; the bridges are simply stuck.

11 Impulse $\rightarrow$ down the tubules $\rightarrow \text{Ca}^{2+}$ floods out $\rightarrow$ troponin/tropomyosin $\rightarrow$ cross-bridges, force rises $\rightarrow$ peak force $\rightarrow$ SERCA pumps back $\rightarrow$ relaxed. Force lasts longest by far (≈ 100 ms vs ≈ 1 ms for the impulse); that is why a second impulse can land while force is still up.

12 Second impulse the same; force bigger ($\approx 1.8\times$ a twitch). Impulses do not add — force does, because Ca^{2+} from the second release lands on top of what SERCA had not yet cleared. At a high rate it is Ca^{2+} removal that runs out of time, so troponin stays switched on: fused tetanus. Holding a cup: tetanus (unfused or partly fused). Nobody produces a single twitch voluntarily.

13 Well past threshold (about $2\times$ what is needed: the safety margin). Half block: the same. A command fails only when the end-plate potential falls short of threshold — most receptors gone (myasthenia), a deep enough block (rocuronium), or the terminal's ACh running down.

C Many — a continuous barrage, not one spike: motor units firing at ~ 10 –50 Hz; every voluntary movement is a tetanus (scene 12). Harder squeeze: both taller (more, larger units recruited) and busier (higher rate). Say explicitly that the EMG is the summed signal of thousands of fibres firing out of step, recorded from outside — not the single-fibre membrane potential on the sim's trace.